8.8 Infinite graphs as limits of finite ones

In the last section we saw how the space $|G|$, for a locally finite graph G, seems to appear as a 'limit' of the finite minors G_n of G obtained by contracting the components left on deleting its first n vertices. We now make this relationship between $|G|$ and the G_n more formal. Clarifying this can help a lot with transferring theorems for finite graphs to infinite ones – which, after all, is the idea behind considering $|G|$ in the first place.

Let $(P, \leqslant)$ be a *directed* partially ordered set, one such that for all p, q there exists an r such that $p \leqslant r$ and $q \leqslant r$. A subset $Q \subseteq P$ is *cofinal* in P if for every $p \in P$ there exists some $q \in Q$ with $p \leqslant q$. *directed* *cofinal*

For every $p \in P$ let X_p be a compact Hausdorff topological space; later, these will represent finite graphs. Assume that we have continuous maps $f_{qp}\colon X_q \to X_p$ for all $q > p$, which are *compatible* in that, whenever $r > q > p$, we have $f_{qp} \circ f_{rq} = f_{rp}$. The family $\mathcal{X} = (X_p \mid p \in P)$, together with these *bonding maps* f_{qp}, is called an *inverse system*. *inverse system* *bonding maps*

The set X of all $x = (x_p \mid p \in P)$ with $x_p \in X_p$ and $f_{qp}(x_q) = x_p$ for all $p < q$ in P is the *inverse limit* $X = \varprojlim \mathcal{X}$ of $\mathcal{X}$. We give it the subspace topology from the product space $\prod_{p \in P} X_p$ which, like the X_p, is Hausdorff and compact by Tychonoff's theorem. *inverse limit*

The space $X = \varprojlim \mathcal{X}$ is the intersection, over all $q \in P$, of the sets $X_{<q}$ of all $(x_p \mid p \in P) \in \prod_p X_p$ that satisfy $f_{qp}(x_q) = x_p$ for all $p < q$. Using the fact that the X_p are Hausdorff and the maps f_{qp} are continuous, one can show that these subsets $X_{<q}$ of $\prod_p X_p$ are closed. Thus, $X = \bigcap_{q \in P} X_{<q}$ is closed in the compact space $\prod_p X_p$, and therefore compact.

As P is directed, the sets $X_{<q}$ have the finite intersection property, as long as the X_p are non-empty. Then $X = \bigcap_q X_{<q}$ is also non-empty:

Lemma 8.8.1. *$X = \varprojlim (X_p \mid p \in P)$ is a compact Hausdorff space. It is non-empty if $X_p \neq \emptyset$ for all $p \in P$.* □

Given a graph $G = (V, E, \Omega)$, consider as $P = P(G)$ the set of all finite partitions of V with only finitely many cross-edges. Letting $p \leqslant q$ whenever q refines p makes P into a directed partially ordered set. For each p, let G/p be the finite multigraph on p whose edges are the cross-edges of p.[15] The vertices of G/p that are non-singleton partition classes are its *dummy vertices*. The other vertices of G/p, those of the form $\{v\}$, we consider to be vertices of G and refer to them as v. G, V, E, Ω $P(G),\ P$ G/p *dummy vertices*

On the compact spaces $X_p := |G/p|$ we have compatible quotient maps $f_{qp}\colon X_q \to X_p$ for $q > p$ which send the vertices of G/q to the vertices of G/p that contain them as subsets; which are the identity on the edges $X_p,\ f_{qp}$

[15] If the partition classes $U \in p$ are connected in G, then G/p is the minor of G obtained by contracting them. But we do not require them to be connected.

of G/q that are also edges of G/p; and which send any other edge of G/q to the dummy vertex of G/p that contains both its endvertices in G/q. Let

$\|G\|$

$$\|G\| := \varprojlim \,(\, X_p \mid p \in P \,),$$

with these f_{qp} as bonding maps.

Theorem 8.8.2. *If G is locally finite and connected, then $\|G\|$ is homeomorphic to $|G|$.*

Proof. As $\|G\|$ is compact and $|G|$ is Hausdorff, it suffices to construct a continuous bijection $\sigma\colon \|G\| \to |G|$. Let $x = (\, x_p \mid p \in P \,) \in \|G\|$ be given.

If there exists $p \in P$ such that x_p is not a dummy vertex of G/p, then $x_p \in |G| \setminus \Omega$ and we let $\sigma(x) := x_p$. To see that this is well defined, consider two such points x_p and $x_{p'}$ and pick $q > p, p'$. Then x_q is not a dummy vertex either, and $x_p = x_q = x_{p'}$ by the definition of f_{qp} and $f_{qp'}$.

Suppose now that x_p is a dummy vertex for every p. For every $n \in \mathbb{N}$ let S_n be the set of the first n vertices of G in some fixed enumeration, and let $p_n \in P$ consist of the vertices in S_n as singleton partition classes and the vertex sets of the components of $G - S_n$ as the remaining partition classes. This sequence $p_0, p_1, \ldots$ is cofinal in P, since every $p \in P$ is refined by every p_n with n large enough that all the cross-edges of p have their endvertices in S_n.

As $f_{qp}(x_q) = x_p$ whenever $p = p_m < p_n = q$, the connected vertex sets $U_n = x_{p_n}$ form a descending sequence $U_0 \supseteq U_1 \supseteq \ldots$. It is straightforward to construct a ray R in G that has a tail in $G[U_n]$ for every n. Let ω be the end of R.

For every $p \in P$ the set $U = x_p$ contains every U_n with $p < p_n$ as a subset. As the p_n are cofinal in P, every $G[x_p]$ thus contains a tail of R. Conversely, for every end $\omega' \neq \omega$ there is an n such that $G[U_n]$ contains no ray from ω'. Thus, ω is the unique end of G that has a ray in $G[x_p]$ for every $p \in P$. Let $\sigma(x) := \omega$. This completes the definition of σ.

To see that σ is injective, consider distinct points $x, x' \in \|G\|$, differing in their components $x_p \neq x'_p$ say. If p can be chosen so that one of these is not a dummy vertex of G/p, then clearly $\sigma(x) \neq \sigma(x')$. Otherwise $U = x_p$ and $U' = x'_p$ are disjoint vertex sets in G separated by finitely many edges, and $\sigma(x)$ is an end with a ray in $G[U]$ while $\sigma(x')$ is an end with a ray in $G[U']$. Thus again, $\sigma(x) \neq \sigma(x')$.

To see that σ is surjective, let $x \in |G|$ be given. If x is not an end, choose $p(x) \in P$ so as to contain the vertex x, or the endvertices of the edge containing x, as singleton partition classes. For every $q \geqslant p(x)$ in P let $x_q := x$, and for every $p' < q$ for some such q let $x_{p'} := f_{qp'}(x)$. Then $(\, x_p \mid p \in P \,)$ is a well-defined point in $\|G\|$ which σ maps to x.

If x is an end, it has a ray in $G[x_p]$ for exactly one dummy vertex x_p of G/p for every $p \in P$. These satisfy $f_{qp}(x_q) = x_p$ whenever $p < q$, so $(\, x_p \mid p \in P \,)$ is a point in $\|G\|$ which σ maps to x.

Let us show that σ is continuous at every point $x = (\, x_p \mid p \in P\,)$ of $\|G\|$. If $\sigma(x)$ is not an end, there exists some $p(x) \in P$ such that $\sigma(x) = x_{p(x)}$, which is a point in $X_{p(x)}$ but not a dummy vertex. Then every basic open neighbourhood O of $\sigma(x)$ in $|G|$ is also a basic neighbourhood of this same point $x_{p(x)}$ in $X_{p(x)}$. Then the set $\prod_{p\in P} O_p$ with $O_{p(x)} = O$ and $O_p = X_p$ for all $p \neq p(x)$ is a basic open neighbourhood of x in $\prod_p X_p$. Its intersection with $\|G\|$ is an open neighbourhood of x in $\|G\|$ which σ maps to O.

If $\sigma(x)$ is an end, ω say, consider any basic open neighbourhood $O = \hat{C}_\epsilon(S,\omega)$ of ω in $|G|$. Let $p(\omega) \in P$ be the partition of V into the vertex sets of the components of $G - S$ and the singletons in S. Then $V(C)$ is a dummy vertex of $G/p(\omega)$; call it $x_{p(\omega)}$. Let $O_{p(\omega)} \subseteq X_{p(\omega)}$ consist of $x_{p(\omega)}$ and the inner points in O of any C–S edges; these are also points of $X_{p(\omega)}$. As earlier, x has a basic open neighbourhood $\prod_p O_p$ in $\prod_p X_p$ with $O_p = X_p$ for all $p \neq p(\omega)$, whose intersection with $\|G\|$ maps to O under σ. $\square$

Note that our proof did not use that $|G|$ is compact: we reobtain Proposition 8.6.1 as a corollary.

In the proof of Theorem 8.8.2 we found it convenient to work with a cofinal sequence in P instead of the entire set P. This is justified more generally by the following easy lemma:

Lemma 8.8.3. *Let $(\, X_p \mid p \in P\,)$ be an inverse system of compact spaces, let $Q \subseteq P$ be cofinal in P, and consider $(\, X_p \mid p \in Q\,)$ with the same bonding maps. Mapping every point $(\, x_p \mid p \in P\,)$ to its restriction $(\, x_p \mid p \in Q\,)$ then defines a homeomorphism from $\varprojlim\, (\, X_p \mid p \in P\,)$ to $\varprojlim\, (\, X_p \mid p \in Q\,)$.* $\square$

By Theorem 8.8.2 and this lemma, our familiar $|G|$ for locally finite G is the inverse limit of the finite contraction minors G_n of G defined as in Section 8.6. Indeed, for the cofinal sequence $p_0, p_1, \dots$ in P defined in the proof of the theorem, we have $G_n = G/p_n$, and by the lemma $|G|$ is the inverse limit of the corresponding compact spaces X_{p_n}.

Just like $|G|$ itself, every standard subspace X' of $X = |G|$ can be obtained as an inverse limit of finite multigraphs. Indeed, the projections $f_p\colon X \to X_p$ defined by $(\, x_p \mid p \in P\,) \mapsto x_p$ are continuous, so their images $X'_p \subseteq X_p$ of X' are compact since X' is, and the f_{qp} send X'_q to X'_p. Thus, $(\, X'_p \mid p \in P\,)$ is an inverse system with bonding maps $f'_{qp} := f_{qp} \restriction X'_q$, and $X' = \varprojlim\, (\, X'_p \mid p \in P\,)$.

More typically, we would like to *find* a standard subspace X' with certain desired properties – for example, a topological spanning tree. We can then try to construct some X'_p whose inverse limit is X'. It may not be straightforward, however, to find such compatible X'_p for all $p \in P$. Here, Lemma 8.8.3 can help: it is only necessary to find them for all p in some cofinal $Q \subseteq P$. For example, we can construct spanning trees inductively in all the G_n by expanding a dummy vertex in the tree $T_n \subseteq G_n$ to a star in $T_{n+1} \subseteq G_{n+1}$. Then our given bonding maps $X_{p_n} \to X_{p_m}$ will map the subspace X'_{p_n} induced by T_n to that induced by T_m, and these X'_{p_n} will have a topological spanning tree in $X = |G|$ as their inverse limit. This construction is possible only because the partition classes of the p_n are connected in G; we could not perform it on all of $P(G)$.

Arcs and circles in $|G|$, or in a standard subspace, can be obtained easily by applying the following *lifting lemma* with $Y = [0,1]$ or $Y = S^1$. Let $(\, X_p \mid p \in P\,)$ be any inverse system of compact spaces, with bonding maps f_{qp} say, and let X be its inverse limit. Let Y be a topological space with continuous *compatible* maps $g_p \colon Y \to X_p$: maps that commute with the f_{qp} in that $g_p = f_{qp} \circ g_q$ whenever $p < q$. Let us call the family $(\, g_p \mid p \in P\,)$ *eventually injective* if for all distinct $y, y' \in Y$ there exists some $p \in P$ with $g_p(y) \neq g_p(y')$.

compatible maps

lifting lemma

Lemma 8.8.4. *There is a unique continuous map $g \colon Y \to X$ that commutes with the projections $f_p \colon X \to X_p$ in that $g_p = f_p \circ g$ for all $p \in P$. If the g_p are eventually injective, then g is injective.* □

For example, suppose we wish to find an arc in X between some points x and y. We can find a topological x–y path $g \colon [0,1] \to X$ by finding topological $f_p(x)$–$f_p(y)$ paths $g_p \colon [0,1] \to X_p$ that commute with the f_{qp}. If we can make these g_p eventually injective, then g will be injective, and its image will be the desired arc.

Similarly, if we can find compatible circles $g_p \colon S^1 \to X_p$ that are eventually injective, whose images contain all the vertices of G/p, and which commute with the f_{qp}, then g will define a *Hamilton circle* of G, a circle in $|G|$ that traverses every vertex.

Hamilton circle

Exercises

1.$^-$ Show that a connected graph is countable if all its vertices have countable degrees.

2.$^-$ Given countably many sequences $\sigma^i = s^i_1, s^i_2, \dots$ ($i \in \mathbb{N}$) of natural numbers, find one sequence $\sigma = s_1, s_2, \dots$ that beats every σ^i eventually, i.e. such that for every i there exists an $n(i)$ such that $s_n > s^i_n$ for all $n \geqslant n(i)$.

3.$^-$ Can a countable set have uncountably many subsets whose intersections have finitely bounded size?

4.$^-$ Let T be an infinite rooted tree. Show that every ray in T has an increasing tail, that is, a tail whose sequence of vertices increases in the tree-order associated with T and its root.